

Hard questions, and plain answers

Written from the outside: a sceptic who knows lidar, knows Pennsylvania oil and gas, and is not inclined to be generous. Every answer uses the project's own numbers.

Where the honest answer is "we don't know" or "we haven't tested that", it says so. Those are the ones worth rehearsing.

The premise

Q. Pennsylvania has hundreds of thousands of undocumented wells. You surveyed 20 square kilometres. Why should anyone care?

Because the bottleneck is not money, it is addresses. The 2021 Infrastructure Act put \$4.7 billion toward plugging orphaned wells. You cannot plug a well you cannot find, and for most of these there is no record to look up.

This is a method, tested on two tiles. The point is whether the method works, not whether 20 km² is a lot of ground.

Q. Why not just use the state's well records and go look?

That was the first thing tried, and it fails on the data. The DEP list has 20,108 well records for Venango County. **8,054 of them carry a placeholder spud date of 1 January 1800** — the database's way of saying it does not know. Another 1,553 have no date at all. So roughly 48% have no usable date.

Worse, where records exist the coordinates are unreliable. Anything logged before 1990 was read off paper plat maps, carrying 50 to 200 metres of error. On the ground that means the dot and the pit are not in the same place, and sometimes the dot has no pit anywhere near it.

Q. If a well was plugged properly, why does it matter that you found it?

It may not. But "plugged" for a pre-1985 well often meant a wooden plug, or a tree stump, or nothing. Casing was frequently pulled out for scrap, which leaves an open conduit rather than a sealed one. Finding it is step one; whether it needs work is a field decision.

Pennsylvania history

Q. Drake's well was 1859. Why would a 160-year-old hole still leave a mark?

Because what you are seeing is not the hole. It is the disturbed ground around it — the earthen pit dug beside the well for oil and brine, the cleared and levelled pad where the rig stood, and the access track cut in to reach it.

Those are earthworks. In forest, on the Appalachian Plateau, earthworks persist. They fill with leaf litter and silt and become shallower, which is exactly why the measured shape is a dish about half a metre

deep rather than a hole. Measured over the 503 paired pits in 9t: median depth 0.54 m, 15.3 m across the rim, 5.8 m across the floor.

Q. Why 1985? That date looks arbitrary.

It comes from Pennsylvania's Oil and Gas Act. A well counts as orphaned if it was abandoned before the Act took effect, has not been operated by the current owner, and the current owner never took any economic benefit from it.

Before that, walking away was legal. That is the whole reason this category exists.

Q. Your slide says Pennsylvania had 77% of the world's oil. Can you source that?

Have the citation ready before you say it out loud. The defensible version is that Pennsylvania dominated world production for roughly the first two decades after 1859 and remained the leading source into the 1880s. If you cannot cite the exact 77% figure, use the softer claim.

Q. Venango County is Oil Creek and Pithole. Those are 1860s boomtown wells, drilled fast and cheap. Are they even the same kind of feature as a 1920s well?

Probably not, and nothing here has tested that. The annotations were drawn on shape, not on age. A dense 1860s field and a scattered 1920s lease would look different on the ground, and the model was never asked to tell them apart.

It is a fair criticism that the training set mixes eras without saying so.

The lidar itself

Q. Four points per square metre. How do you expect to resolve a half-metre depression with that?

Density is not the limit here — vertical accuracy is, and there is room.

The delivered cloud averages about 4 points per square metre, but what matters is ground returns, and those measure 2.7 per square metre across 9t. A pit 15.3 m across is 184 m², so roughly five hundred ground measurements land inside it. The survey's vertical accuracy requirement is 10 cm RMSE. The feature is 54 cm deep.

So the signal is about five times the noise floor, measured with hundreds of points. That is a workable margin — the difficulty is separating it from other half-metre dishes, not detecting it.

Q. The laser can't see through leaves. How much of the ground are you actually measuring?

Not all of it, and the deck says so. Even after recovering the discarded returns, 8.2% of the ground the laser reached has no ground measurement at all. Those are under canopy thick enough that no pulse reached dirt.

Surveys are flown leaf-off for this reason. It helps; it does not solve it.

Q. You threw out the survey contractor's classification and substituted your own. Why should anyone trust yours over a professional vendor's?

We did not replace it. Where the vendor classified ground, we agree with it and keep it. What we add is ground where the vendor had none.

And the returns we added are not junk. Their measured accuracy was compared against the returns the vendor kept, and both clear the published specification. The vendor discarded good data.

Q. You spend a third of the talk proving the vendor threw away good ground, and then every result you report is built on the vendor's ground. Which is it?

Both, and say so before anyone catches it. The recovery work is a finding about the delivery and a measured opportunity. It is not in the detection pipeline. The derivative builder keeps class 2 only, and the training DEM is identical to the vendor-only DEM cell for cell.

There is a run with the recovered ground in it — it scores 0.915 against the vendor-ground run's 0.928 — and it is not the number in this deck. Quoting the reprocessed result while the pipeline uses the vendor's would be the dishonest version of this answer.

So: the section says what the delivery cost, and the results say what the method achieves on the delivery as it stands. Putting the two together is the next piece of work, not a claim being made today.

Q. Discarded why?

Every return past 18 degrees of scan angle was deleted before delivery. Not degraded gradually — deleted, at exactly 18 degrees.

A clean vertical cliff at a round number is a processing decision, never physics. And it is not universal: that cut appears in six of seven Venango tiles from the March 2020 block and zero of seven McKean tiles from a different block.

Q. What did that actually cost you?

13.8% of the ground the laser reached had no ground measurement under it. Putting the discarded returns back brings that to 8.2% — 2.9 million of 7.1 million cells filled, two holes in five.

Say "the ground the laser reached" and not "the training area", because the denominator matters: 51,470,048 cells caught a return of some kind, and 7,098,612 of those caught no ground return. Against all 81 million cells in the 9t box at 0.5 m it would be 8.8%, and the missing 29.5 million were never sampled at all — ground returns sit 0.61 m apart and the grid asks every 0.5 m, which is our choice of cell size, not the vendor's doing.

On a single cross-section it is starker: 32% of one line across the ground had nothing beneath it, down to 2% with the returns restored.

The artefacts

Q. You found stripes in your own data that are the same size as the thing you are looking for. How do you know your detections are not stripes?

That is the right question, and it is why the artefact is in the deck rather than hidden.

Two different stripe problems were found. On canopy height they are empty cells — no return came back, nothing to draw — and coarsening the grid from 0.5 m to 1 m removes them completely, 3.22% of a test window down to 0.00%.

On the shape layers they are real measurements that disagree with each other. The ground surface has **no empty cells at either resolution**, so those cannot be missing data. They run at 78°, the bearing of the scanner's own lines.

Their amplitude is under a centimetre, measured two ways: 0.91 cm across the nine rows drawn by hand, 0.70 cm on the cross-strike profile. Against a 54 cm pit that is seventy times shallower, so depth is not where the risk lives.

What they share with a pit is spacing. A crest every 3.35 m is the width of the rims we look for, and every layer the model reads — local relief, openness, hillshade — measures shape rather than height, so a ripple this shallow still prints on them. The mitigation is geometry: a pit is a closed, roughly circular dish and a stripe is a long straight line, and the model sees shape over a 128 m patch rather than one cell at a time.

If an earlier version of this answer said "tens of centimetres", that was wrong by a factor of about thirty and is corrected here.

Q. "Mitigation" is doing a lot of work there. Have you measured that the detections are not artefact-driven?

Not directly. That test has not been run. The honest position is that the detections behave like pits — closed, circular, in the right size range, in plausible places — and that the artefact has a different geometry. That is an argument, not a measurement.

Q. You tried three explanations for the stripes and all three failed. Doesn't that mean you don't understand your own data?

It means we do not understand that artefact. A washboard from line-to-line height offsets, a scan-angle or attitude effect, and interswath disagreement were each tested and each refuted by the measurement meant to confirm it.

The data provider's own position is that it is in the raw data and cannot be fixed by the end user. Their suggested fix — averaging points in a radius instead of triangulating — made the shape layers visibly worse, because averaging smooths heights and roughens the slopes between cells.

So: managed, not solved. Saying otherwise would be untrue.

The ground truth

Q. One person drew every label. Then that person's model was scored against that person's drawings. What is precision actually measuring?

Agreement with one annotator. Not correctness.

Precision runs 0.59 to 0.69 on the two tasks that report it — 0.587 for pads and 0.633 for pits under the recall-weighted rule, 0.686 for pits under the balanced one. Roads report correctness instead, 0.828 on the second tile. That is the rate at which the model agrees with one person's judgement about what counts as a pit, on ground that person had already looked at.

A second annotator on a subset is the only thing that measures this, and it has not been done. It is on the next-steps slide for that reason.

Q. Recall 0.928 sounds excellent. Recall of what?

Of pits that were drawn. If there are pits on that tile nobody noticed, they are not in the denominator, and the model is not penalised for missing them.

So 0.928 means: of the 503 pit floors drawn inside 9t, five-fold cross-validation matched 467, and every one of those was scored by a model that had never seen it. It does not mean 92.8% of the wells that are there. (712 is the count of floors drawn across every area; 503 is the 9t subset the models were scored on.)

Q. So you have never confirmed a single well in the field.

Correct, and the deck states it. No field visit, no records check. Everything produced here is a candidate.

That is the single largest gap in the work, and it is the first item on the next-steps slide — because field-checking a sample is what converts recall from a rate against drawings into a rate against reality.

Q. Why did you draw roads and streams at all? You are looking for wells.

Roads because a well that nobody recorded still had to be reached, so an overgrown access track is evidence. 535.9 km of it was drawn by hand.

Streams because the model kept calling them roads. Drainage is not a target — it is a class the model is taught in order to reject it, which worked.

The model

Q. Why a neural network? A pit is a circular depression. Why not just look for circular depressions?

That was the earlier approach — match a template of a mean pit across the terrain layers — and it is not in this deck any more.

The reason is on the slide about layers: no single terrain measure separates a well pit from a natural hollow. Local relief and topographic position light up both. What separates them is a combination, and a

network is a practical way to learn a combination.

Q. You tested four architectures and the biggest one gained 0.002 on pits. Doesn't that mean the network is not doing much?

It means architecture is not the constraint. Three times the parameters and ImageNet pretraining bought a gain smaller than the spread between folds.

The useful reading is that the limit is upstream — the channels the model gets, and the labels it is trained on. That is why the next-steps slide is about channels and labels rather than about bigger models.

Q. Your model sees seven terrain layers and nothing else. You built a canopy height model and an intensity layer and feed it neither. Why?

No good reason. Both are built and neither reaches any detector, so the network only ever sees shape — never material, never vegetation.

That is listed as one of the next changes precisely because it is cheap: a channel, not a new architecture.

Q. How do you know the model didn't just memorise the tile?

Because of how the split was made. The tile is cut into a 12 by 12 grid and whole blocks go to training or testing. Individual pits are never split.

If you split pit by pit, a training pit ends up 30 metres from a test pit, on the same hillside, in the same light. The model would recognise the hillside and score beautifully while learning nothing.

And the second tile is the harder check: 1.5 km away, never seen, thresholds frozen. Recall drops 0.017.

The numbers

Q. "Flags 0.21% of the tile" sounds tiny. What is it in the field?

The tile is 4.5 km square, so 20.25 km². 0.21% of that is about 42,500 m².

Know which run that figure comes from before you quote it beside a recall. It is the threshold sweep at 0.20, which draws 1,041 polygons covering 4.33 ha and recovers 126 of the 127 pits in the held-out blocks. The cross-validation is a different measurement on a different denominator.

Take the cross-validation for the field arithmetic, because it covers the whole tile: to recover 467 real pits the model points at 738 places, so 271 are wrong. A crew checks about seven hundred spots to find just under five hundred pits.

That is a workable ratio. It is not "the computer found the wells".

Q. And the pad model?

Much worse on burden. Recall 0.912 and precision 0.587, and the threshold sweep flags 9.8% of the tile — 197.95 ha, about 2.0 km² to review — at the same recall.

Its finding rate is fine. The amount of ground it asks someone to walk is the problem, and the deck says so rather than reporting only the recall.

Q. Precision 0.63 means a third of your detections are false. On 350,000 wells that is an enormous amount of wasted field time.

Two things. First, that third is measured against one person's drawings, so some of those "false" detections may be real pits the annotator missed. Nobody knows which, because nothing has been field-checked.

Second, the threshold is adjustable. Raising it trades recall for precision. The right setting depends on whether the cost is a wasted site visit or a missed well, and that is a programme decision, not a modelling one.

Generalisation

Q. Two tiles, 1.5 km apart, same county, same survey, same flight block. What exactly has been generalised?

Nothing, and the conclusion slide says so in those words.

What the second tile shows is that the model did not memorise the first one — recall fell from 0.928 to 0.911 on ground it had never seen, with thresholds frozen. That is a real result and a narrow one.

A tile from a different acquisition is the actual generalisation test, and it has not been run.

Q. Road recall on the first tile is 0.982 and on the second it is 0.759. That is not a small drop.

They are not the same measurement, and the second one is deliberately the hard half. On the second tile the truth splits in two: 138.23 km is a previous model's output that a human vetted, and 48.87 km was drawn fresh by hand. Every model scores 0.96 or better on the vetted half, because it is scoring against something descended from a model. It ranks nothing, so only the hand-drawn half is reported: 1,162 of 1,496 chunks, 0.777 by chunk and 0.759 by length, at threshold 0.50.

Against the whole 187.1 km of truth on that tile the same model reads 0.915. The deck quotes the lower number on purpose.

Q. Your map of the second tile shows 42 pit candidates, but you also say the model found 139 of 153 pits there. Those cannot both be true.

They are two different models and the slide now says so. The 42 pit and 161 pad candidates are the earlier, pre-U-Net detector's output from June, and that is the layer the map draws. The 139 of 153 is the U-Net transfer, five folds with thresholds frozen from the first tile, which predicts between 363 and 1,162 polygons per fold.

The map has not been redrawn from the U-Net. That is a presentation gap, not a disagreement about a result.

Q. You report no precision at all on the second tile. Convenient.

It is not computable there and the reason is structural. Only part of that tile was ever annotated: the drawn extent covers 70.3% of it, only a third of its blocks hold a drawn pit, and the largest single unswept region is 63% of the tile. An unmatched prediction inside unswept ground may be a false positive or a pit nobody drew, and nothing in the data separates the two.

Reporting a precision figure there would be a number with no meaning attached. Finishing the annotation is on the next-steps slide for exactly this reason.

Q. If the artefacts are specific to one flight block, does your method only work on that block?

The opposite is more likely. McKean County is QL1 — roughly twice the point density — and shows none of the 18-degree cut in seven tested tiles. Cleaner data should make this easier, not harder.

But that is an expectation, not a result. The model has not been run there.

The questions with no good answer yet

Q. Pennsylvania forests are full of relict charcoal hearths — flat circular platforms 10 to 15 metres across, cut for the charcoal iron industry, and mapped in their tens of thousands elsewhere in the state with exactly this kind of lidar. Your pit rims average 15 metres across. How do you tell them apart?

This is the sharpest question in the set, and the honest answer is that it has not been addressed. Charcoal hearths appear nowhere in the annotations, the negative classes, or the analysis.

The argument that they should separate is geometric: a hearth is a **flat or slightly raised platform**, often cut into a slope, while a well pit is a **closed concave dish**. Negative openness — the layer that matters most for pits — measures enclosure, and a platform is not enclosed. On a slope, though, a hearth's back-cut can read as a partial depression, and that is where confusion would live.

Venango borders Clarion County, which was a genuine charcoal-iron district, so hearths are plausibly present. Nobody has checked.

If this comes up: concede it, say the geometry argues for separation, and say it needs labelling and testing. Do not claim it is handled.

Q. What else looks like a well pit?

Several things, and only some are covered:

- **Vernal pools and seasonal ponds** — natural, closed, and the right size. Ponds are listed as a proposed negative class and are not yet trained in.

- **Cellar holes** from abandoned farmsteads. These turned up in the annotation quality-control review as unusual-but-correct features, so they are already a known source of confusion.
- **Tree-throw pits** from windthrown trees. Smaller, typically 1 to 3 metres, and paired with an adjacent mound — usually separable on size.
- **Strip-mine scrapes and quarry workings.** Listed as a proposed negative class.
- **Sinkholes** are the one thing you can dismiss cleanly: this is Appalachian Plateau sandstone and shale, not carbonate karst.

Q. How clean is the training data? Has anybody checked the labels?

Not the labels these models were trained on, and do not claim otherwise.

There was an annotation quality-control pass — 856 measured pits, 90 flagged as unusual, the top 30 reviewed, about half of those genuine mis-clicks. It was pulled from the deck on purpose. It ran in April 2026 against a different annotation set of 861 pits, over tiles 9t/mk5/mk/mkf, and the annotations were rebuilt from 527 to 712 features on 4 September with the split reassigned. So it predates everything in this deck and describes labels no model here saw.

If the question comes up: say a QC pass was run on an earlier annotation set and found an error rate of roughly that order, that it has not been repeated on the current set, and that a second annotator on a subset is the item on the next-steps slide that would actually settle it. Do not quote 5% as a property of the training data.

Practical

Q. What would it take to run this on a whole county?

The processing is not the hard part — it is reproducible and the derivative build is scripted. Two things scale badly.

The first is annotation. Every label in this project was drawn by one person by hand.

The second is verification. Without field checks, running it wider produces more candidates of unknown reliability, which is not obviously useful to anyone.

Q. What is the single biggest weakness?

No field validation. Every number in the deck is measured against hand-drawn annotation by one person. Until somebody walks to a sample of candidates and reports back, "recall 0.928" means agreement with a drawing, not with the ground.

Q. If you had to defend one number, which is it?

Pits: 0.911 on a tile the model had never seen, with the thresholds frozen from the first tile and nothing retuned, against 0.928 at home.

If you want the burden figure beside it, quote it from its own run rather than bolting it onto the cross-validation: the threshold sweep at 0.20 finds 126 of the 127 pits in the held-out blocks while drawing 1,041 polygons over 4.33 ha, 0.21% of the tile.

The transfer number is the one that matters. Detection rates are easy to inflate by flagging everything, and easy to inflate again by tuning on the ground you report. Finding nearly all of the features on a tile the model had never seen, with nothing retuned, is the result.